INDUSTRY PROJECT: PROJECT REPORT

Title : Employee Attrition Analysis using Power BI

Name of the Company : TCS iON

Name of the Institute : Gurunanak Institute of Technology

Project Duration : 18 - 04 - 2026 to 22 - 04 - 2026

Submitted By : Sonu Dholpuriya

Total Effort (hrs.) : 45 Hours

Project Environment

Jupyter Notebook

Windows OS

Power BI

VS

Tools Used

Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn

Power BI
Excel / CSV
SQL

Acknowledgement

I would like to thank my institute and mentors for providing guidance and support to complete this project successfully.

Objective & Scope

To analyze employee data and identify factors affecting attrition and performance, and build a model to predict employee turnover.

Problem Statement

Employee attrition leads to increased cost and productivity loss. The goal is to analyze and predict employee attrition using data.

Existing Approaches

Traditional methods rely on manual analysis and assumptions, which are time-consuming and less accurate.

Approach / Methodology

- Data Cleaning and Preprocessing
- Handling Missing Values
- Outlier Detection (IQR)
- Data Visualization (EDA)
- One-Hot Encoding
- Train-Test Split
- Logistic Regression Model

Workflow

1. Data Collection
2. Data Cleaning
3. Exploratory Data Analysis
4. Feature Encoding
5. Model Training
6. Model Evaluation

Assumptions

- Data is reliable and correctly represents employee behavior
- Missing values handled appropriately
- Attrition is binary (0/1)

Implementation

- Cleaned dataset by handling missing and inconsistent values
- Visualized data using charts (histogram, bar, scatter, heatmap)
- Applied encoding for categorical data
- Trained Logistic Regression model
- Evaluated using accuracy, classification report, and confusion matrix

Solution Design

A classification model is used to predict whether an employee will leave the company based on input features.

Challenges & Opportunities

- Data imbalance issue in attrition
- Model bias initially
- Opportunity to improve using advanced models

Reflection on the Project

This project helped in understanding data cleaning, visualization, and machine learning concepts in real-world scenarios.

Recommendations

- Improve employee salary structure
- Focus on departments with high attrition
- Improve employee engagement and work environment

Outcome / Conclusion

The project successfully identified key factors affecting attrition and built a predictive model for employee turnover.

Enhancement Scope

- Use advanced models (Random Forest, XGBoost)
- Apply feature scaling
- Build dashboard for visualization

Link to Code

https://github.com/sonu123-sd

References

- Scikit-learn Documentation
- Pandas Documentation
- Online resources

Research Questions

- What factors affect attrition?
- How does salary impact retention?

References

- Scikit-learn Documentation
- Pandas Documentation
- Online resources
- Power BI Documentation
- Python Libraries Documentation